Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti |

AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Backend AI engineer with **security-first development background** - built AgentProbe's AI compliance eval engine (prompt injection detection, tool misuse scoring, hallucination flagging) and implemented PHI encryption + HIPAA compliance at scale at Entrata. 8+ years production Python; designed and deployed secure backend systems handling 100K+ users across financial and healthcare data. AgentProbe is an open-source LangGraph agent platform with multi-provider LLM routing, RAG pipeline, and a 5-metric safety/compliance evaluation engine. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

AI Security & Safety Evaluation | Prompt Injection Detection | LLM Systems & Agents (LangChain/LangGraph) | Backend Infrastructure (Python) | Secure Coding & PHI/HIPAA Compliance | LLM-as-Judge Compliance Scoring | FastAPI / AWS / Docker | Multi-Provider LLM Routing | RAG + Vector Search | CI/CD (GitHub Actions) | Production at Scale (1M+ records) | Cross-Functional Delivery

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Achieved **100% PHI compliance** by designing and implementing Libsodium encryption for all resident health data, migrating client documents to AWS S3 with automated secure pipelines - **zero security incidents** across 100K+ monthly active users.
- Applied **Python, PyTorch, and OpenAI API** to automate HUD income limit data transformation across **1.6M financial records** - secure handling of regulated financial data at production scale.
- Built and delivered **20+ full-stack modules** (Python, JavaScript, PostgreSQL) across **120+ Agile sprints**; enforced secure coding standards across all major releases; conducted peer code reviews with security focus.
- Improved production backend performance by **30%**; maintained **99.5% uptime** for 6+ consecutive months via TDD/BDD; built **5+ internal debugging tools** adopted across multiple squads.
- Led Docker-based containerization for 30%+ of microservices; reduced technical debt by resolving **100+ high-priority bugs**.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Mentored 30+ students in Python and databases; translated complex systems concepts across technical and non-technical audiences.

PROJECTS

AgentProbe: AI Safety & Compliance Evaluation Platform OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Production LangGraph agent platform with a **5-metric AI safety and compliance evaluation engine** - directly applicable to AI security infrastructure. **Compliance & safety scoring**: LLM-as-judge evaluating prompt injection risk, policy violations, hallucination, tool misuse, and retrieval compliance across every agent run. **Regression tracking**: anomaly detection flags safety regressions before they reach users. **CI/CD safety gates**: eval pipeline blocks builds on compliance failures. **Multi-provider LLM routing**: Anthropic, OpenAI, Groq, Ollama with fallback guardrails. **Full-stack**: 15-endpoint FastAPI backend + React observability dashboard. Built greenfield, end-to-end, as open-source. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: Secure NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

NLP pipeline mapping natural language queries to FHIR APIs via LLMs, BERT, and embeddings. **HIPAA-compliant RBAC** - role-based access controls for PHI. Demonstrates building AI systems with security and compliance built-in from architecture. **~60% retrieval time reduction.** *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection (RF vs LR) → rigorous evaluation (MDI + Permutation Importance) → deployment. **85%+ accuracy.** End-to-end applied ML with systematic measurement and deployment.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

AI & Security: LangChain, LangGraph, LLM Safety Evaluation, Prompt Injection Detection, LLM-as-Judge Compliance Scoring, Multi-Provider LLM Routing (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, Tool Misuse Detection, FastAPI

Security & Compliance: Libsodium Encryption, PHI/HIPAA Compliance, Secure Coding Standards, AWS S3 Secure Pipelines, RBAC, Zero-Incident Production Systems

ML & Data: PyTorch, scikit-learn, BERT, spaCy, NumPy, Pandas, Feature Engineering, Statistical Modeling

Cloud & DevOps: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), PostgreSQL, MySQL, RabbitMQ

Process: TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum